



VIMANSH MAHAJAN

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EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi) - Bachelor of Technology (B.Tech.) in Computer Science Engineering (CSE) - Dean's List (2024–25) • Flipkart GRiD 7.0 National Semi-Finalist • Adobe GenSolve Top 5%	CGPA: 9.28 / 10.00	202 - 2026
The Vivekanand School, Narela, New Delhi JEE Mains 2022 – 99 Percentile (AIR 9,200) • JEE Advanced 2022 – Qualified (AIR 10,109)	CBSE XII: 97.2%	2022
Delhi Public School, Noida	CBSE X: 96.8%	2020

EXPERIENCE

- Bank of New York (BNY) | Software Engineer Intern** May 2025- July 2025
- Architected a production-grade content automation system using **Spring Boot, Python LLM services, Oracle DB, and Angular**, transforming raw finance notes and images into publication-ready market commentary.
 - Reduced content turnaround time from **4–5 hours to under 10 minutes** by automating structured note capture, AI drafting in author style, contextual image insertion, and final publishing workflows.
 - Containerized backend services and deployed production workloads on **Google App Engine**, improving portability and release efficiency.
- Complex Systems (CoSy) Lab | UG Researcher** Jan 2025- May 2025
- Developed **RecipeGPT**, a generative NLP system trained on the **Recipe1M** corpus, delivering high-quality recipe generation with **BERTScore > 0.85** and **ROUGE-L 0.32**.
 - Built a custom **NER-driven preprocessing pipeline** to normalize ingredient inputs and improve model conditioning on structured entities.
 - Designed a **3-stage GPT-2 fine-tuning pipeline** over **100k+ recipe instructions**, progressively training on instructions, ingredient-title conditioning, and NER-enhanced tokenized inputs with label masking.

SKILLS

Languages: Python, Java, C++

Frameworks & Libraries: Spring Boot, React, Angular, Hugging Face Transformers, PyTorch

Databases & Tools: PostgreSQL, MySQL, Oracle DB, Git, Docker, REST APIs

Core CS: Data Structures & Algorithms, OOP, DBMS, Computer Networks, System Design

AI / ML: Machine Learning, NLP, LLM Fine-Tuning, Model Evaluation, Data Science

PROJECTS

- OptiWealth: Quantitative Portfolio Optimization Platform** | *Spring Boot, Python, yfinance, React, PostgreSQL* Nov 2025
- Architected a multi-service portfolio analytics platform with **React frontend, Spring Boot REST backend**, and Python compute services for quantitative workloads.
 - Built secure end-to-end workflows using **Spring Security + JWT**, enabling authenticated portfolio creation, tracking, and optimization.
 - Implemented portfolio analytics covering returns, volatility, Sharpe ratio, **VaR/CVaR**, beta, drawdown, and diversification across **NSE-listed equities**.
 - Developed forecasting and optimization engines using **ARIMA, GARCH, Monte Carlo simulation, efficient frontier**, and **max-Sharpe/min-volatility** strategies, with AI-generated plain-language investment insights.
- Healthcare Answer Summarization** | *Python, PyTorch, Hugging Face Transformers, LoRA* Apr 2025
- Engineered a parameter-efficient **BART-large-CNN + LoRA** pipeline for multi-perspective summarization of healthcare Q&A content across five response intents.
 - Architected a **dual-head classifier-generation system** and improved robustness via selective fine-tuning on low-quality generations.
 - Surpassed a **Flan-T5 benchmark** by +3.24 BLEU and +0.0355 BERTScore, reaching **0.8782 BERTScore**.
- OncoHelp-AI: Medical Decision Support for Brain MRI** | *Python, Streamlit, Grad-CAM, LLMs* Dec 2025
- Architected a multi-stage **brain MRI decision-support system** using ResNet18 and EfficientNet models for image validation, tumor screening, and subtype classification.
 - Integrated **Grad-CAM heatmaps** and **Monte Carlo Dropout**-based uncertainty estimation to surface interpretable predictions with calibrated confidence signals.
 - Developed a safety-constrained LLM assistant using **Gemini** and **BioGPT** for conservative summaries and scoped follow-ups.

ADDITIONAL INFORMATION

- Coding Platforms:** Leetcode (300+ problems solved), Codeforces (*Pupil- Max Rating: 1275*)
- Certification:** Quantitative Finance & Algorithmic Trading in Python
- Hobbies:** Debate, Fitness, Badminton
- Volunteering:** Volunteer, Seva Sankalp NGO